

## Training Configuration and Summary:

Dataset used: Training: rssi\_odom\_dataset\_11apr25\_isolado.parquet | Experiment: rssi\_odom\_dataset\_12apr25\_isolado.parquet

model\_type: LSTM  
group\_sec: 100  
timesteps\_orig: 60  
batch\_size: 1024  
min\_units: 16  
max\_units: 512  
units\_step: 16  
dropout\_min: 0.2  
dropout\_max: 0.3  
dropout\_step: 0.1  
activation\_functions: ['tanh', 'relu', 'elu']  
optimizers: ['adam', 'rmsprop', 'Nadam']  
min\_layers: 1  
max\_layers: 2  
max\_epochs: 150  
hyperband\_iterations: 3  
group\_msec: 100  
samples\_per\_sec: 10  
forecast\_horizon\_sec: 4  
forecast\_steps: 40  
timesteps: 60

### Dataset Parameters:

Training:  $\Delta t \approx 12000.0$  ms | samples/window=1.00 | timesteps=60

Experiment:  $\Delta t \approx 12000.0$  ms | samples/window=1.00 | timesteps=60

### Tuning:

Trials executed: 10/10

### Training Performance:

Loss: 0.0033 | MAE: 0.0445

### Validation Performance:

Loss: 0.0102 | MAE: 0.0793

## LSTM Multi-Step Training Report

Report Generated: 2025-08-04 11:15:08

Training Start: 2025-08-04\_10-33-36

Training End: 2025-08-04\_10-56-55

Training Duration: 23.32 minutes

Predictions Start: 2025-08-04\_10-56-55

Predictions End: 2025-08-04\_11-00-37

Predictions Duration: 3.69 minutes

### Best Hyperparameters:

Number of Layers: 2

Activation: relu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 3.3514

MSE: 17.7869

RMSE: 4.2174

R2 Score: 0.4748

Dataset size before down-sample: 31,097 rows  
Dataset size after down-sample: 3,738 rows

First 5 timestamps BEFORE:

1744464597710  
1744464597720  
1744464597744  
1744464597757  
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366  
1744464971377  
1744464971390  
1744464971402  
1744464971414

First 5 timestamps AFTER:

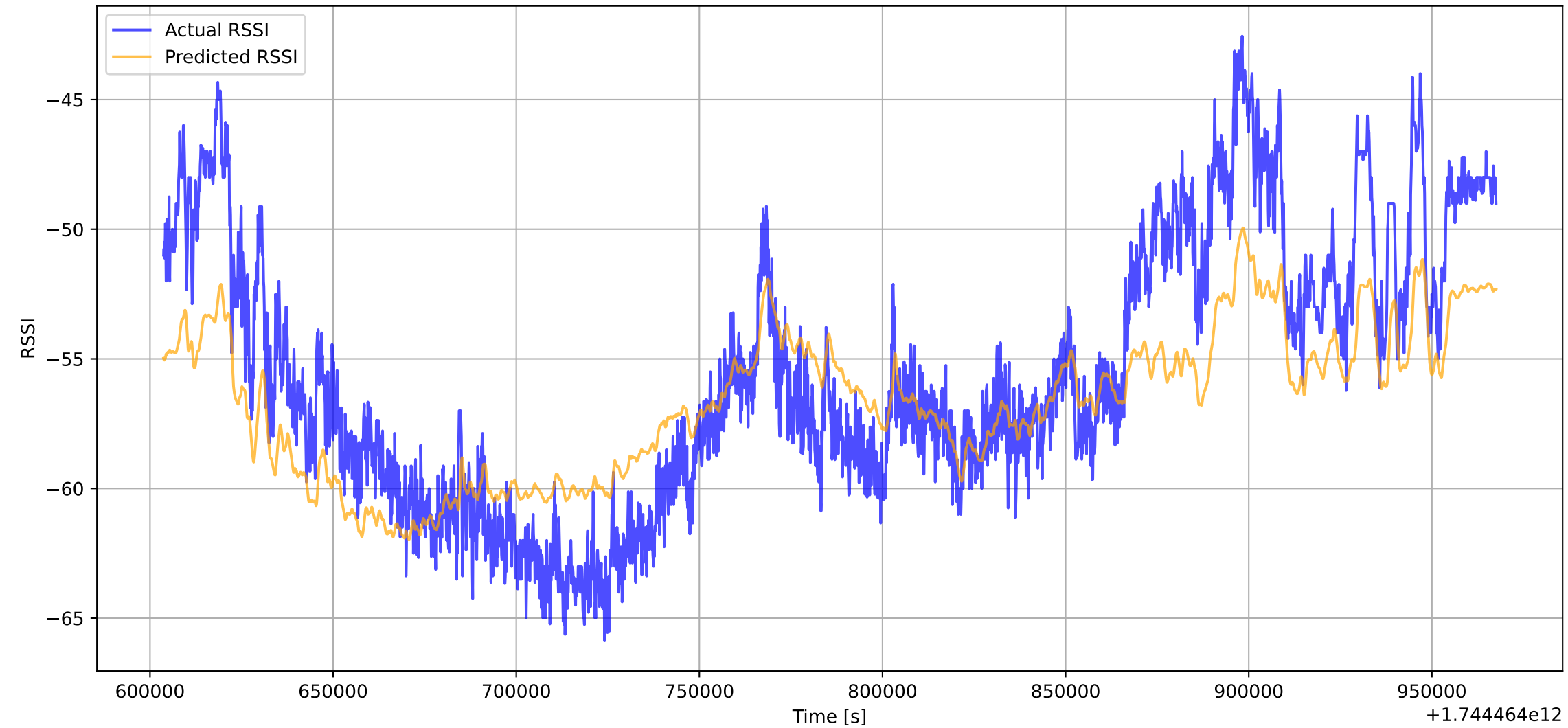
1744464597710  
1744464597812  
1744464597900  
1744464598008  
1744464598103

...

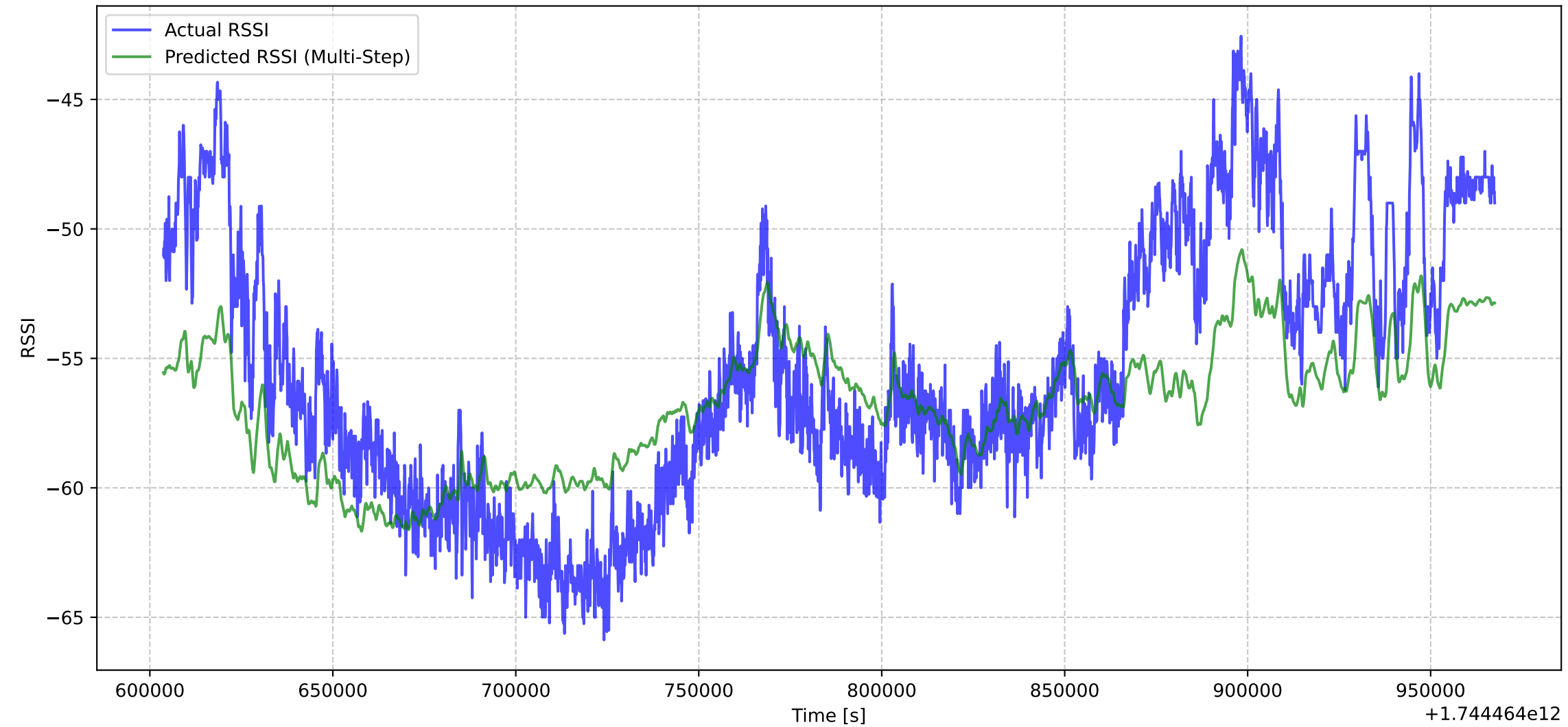
Last 5 timestamps AFTER:

1744464971004  
1744464971101  
1744464971209  
1744464971306  
1744464971402

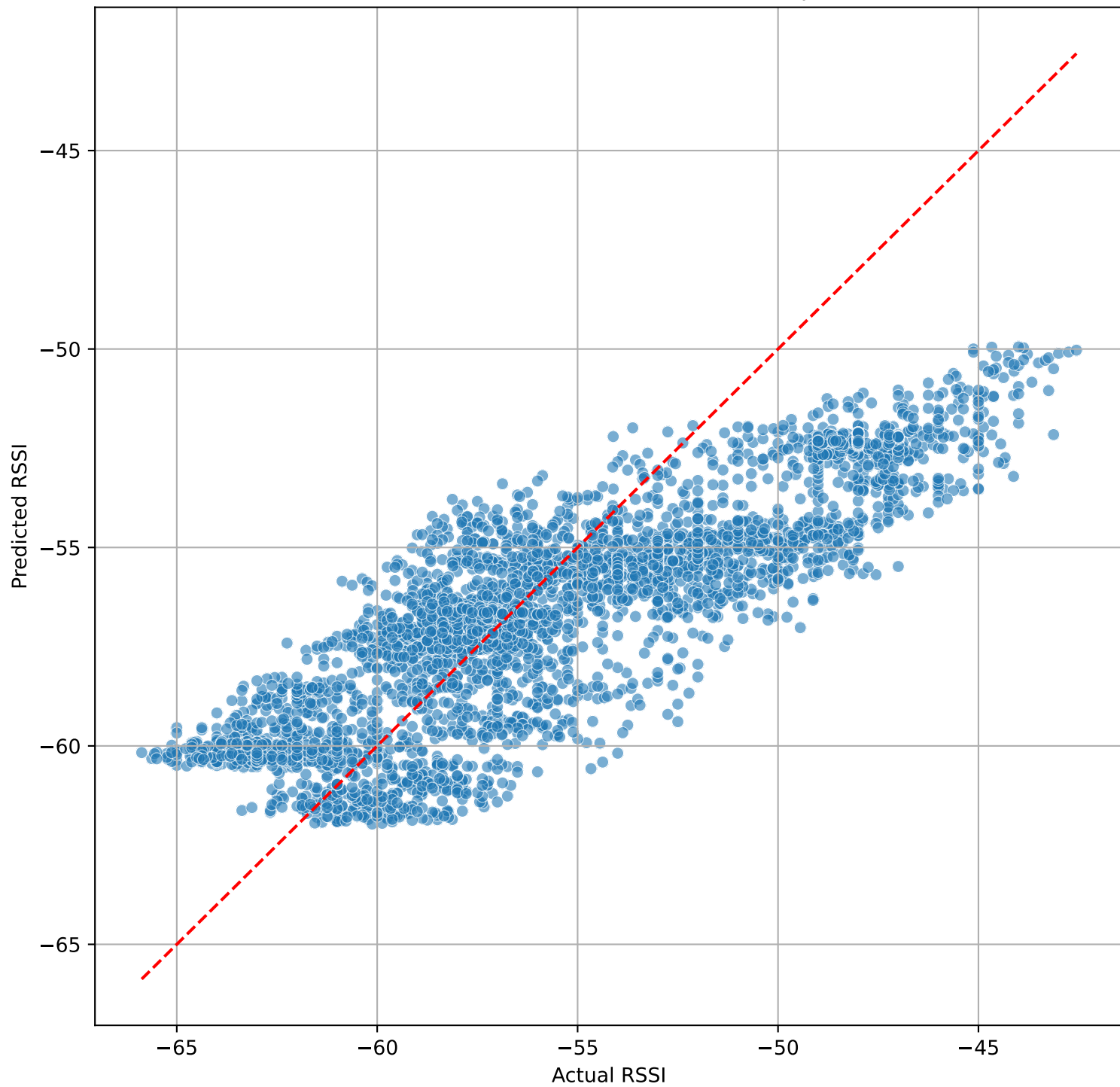
Actual vs Predicted RSSI (First Step)



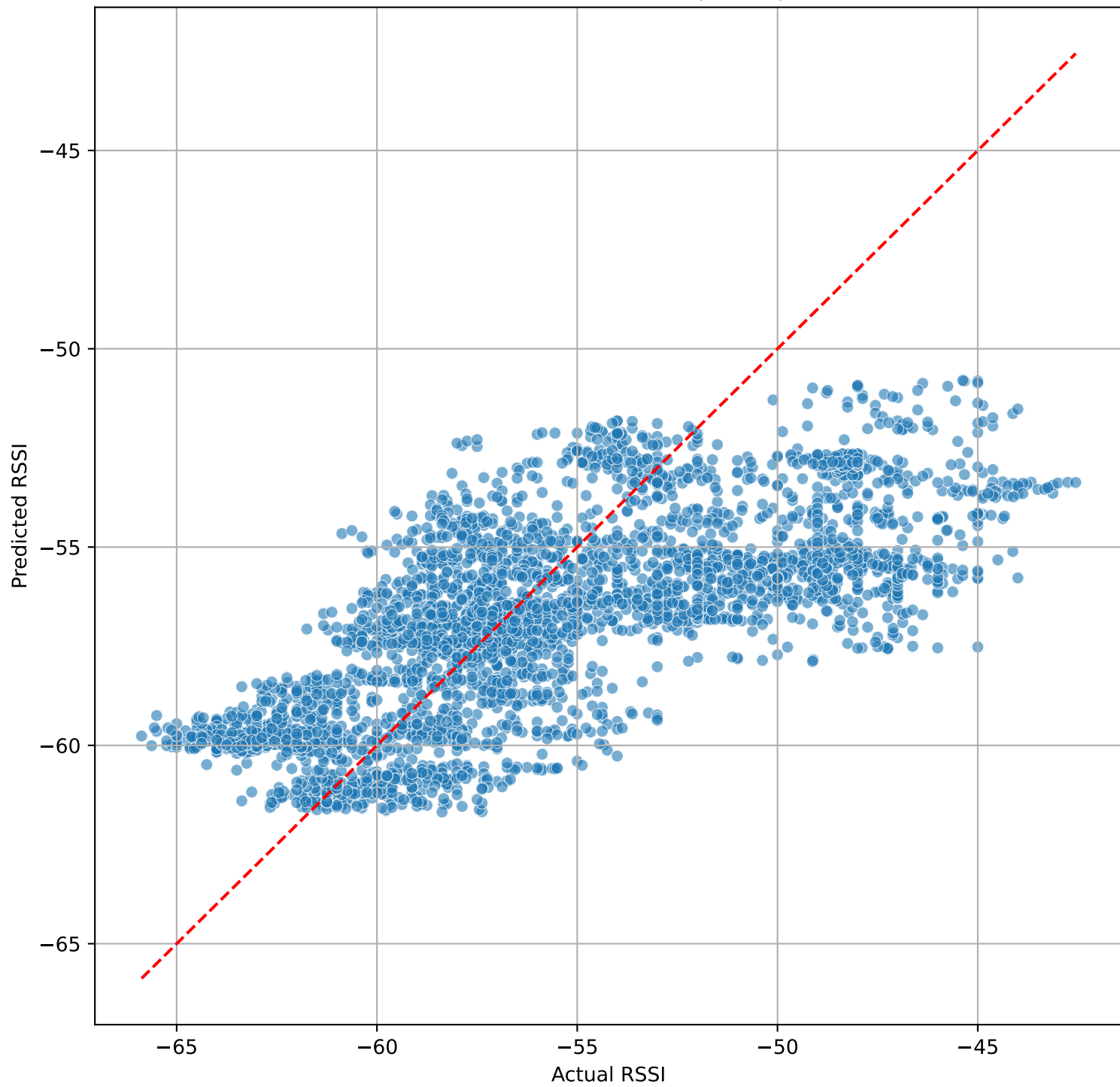
Actual vs Predicted RSSI (Multi-Step (step 40 $\approx$  4.0 s))



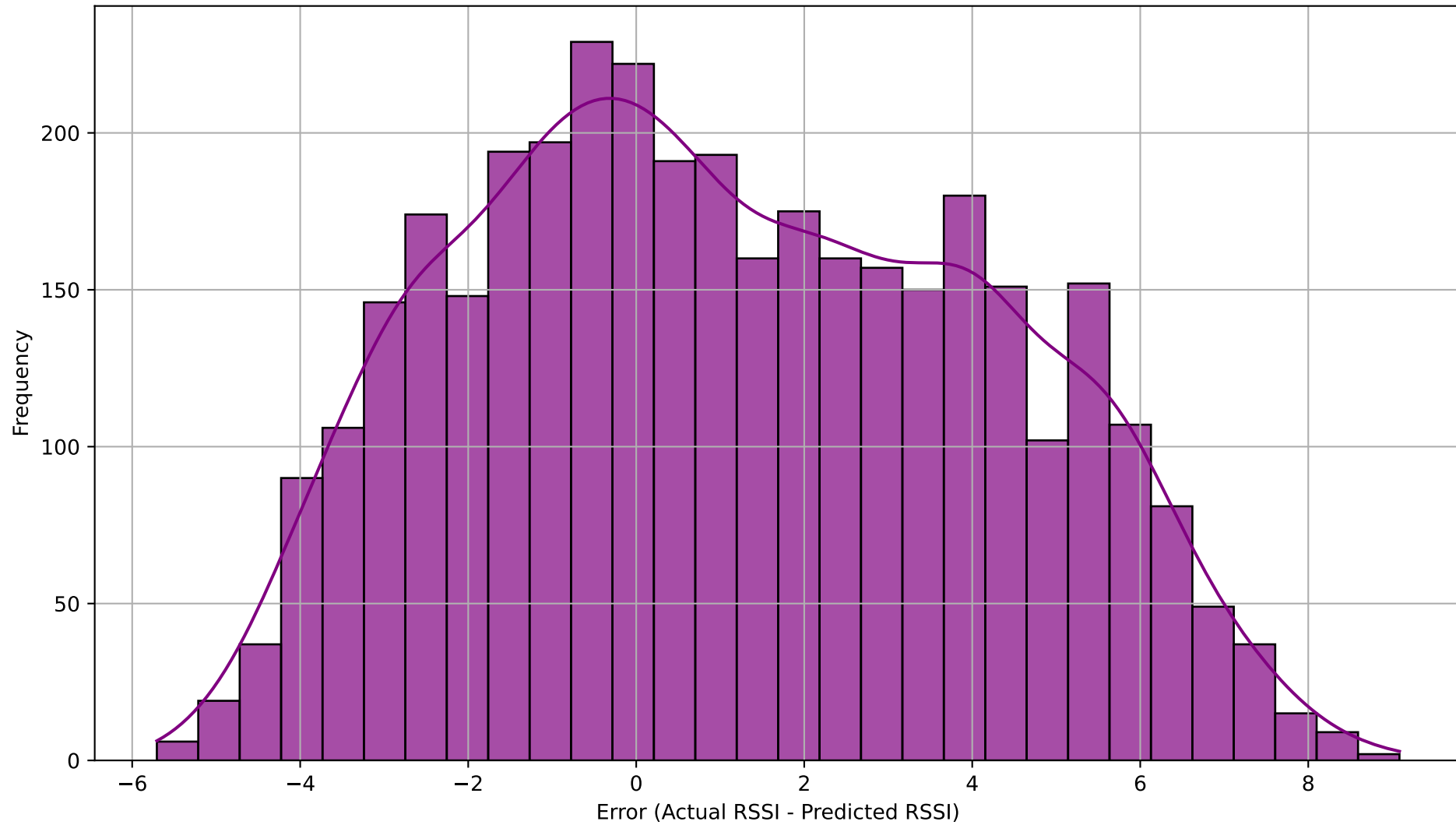
Actual vs Predicted RSSI (First Step)



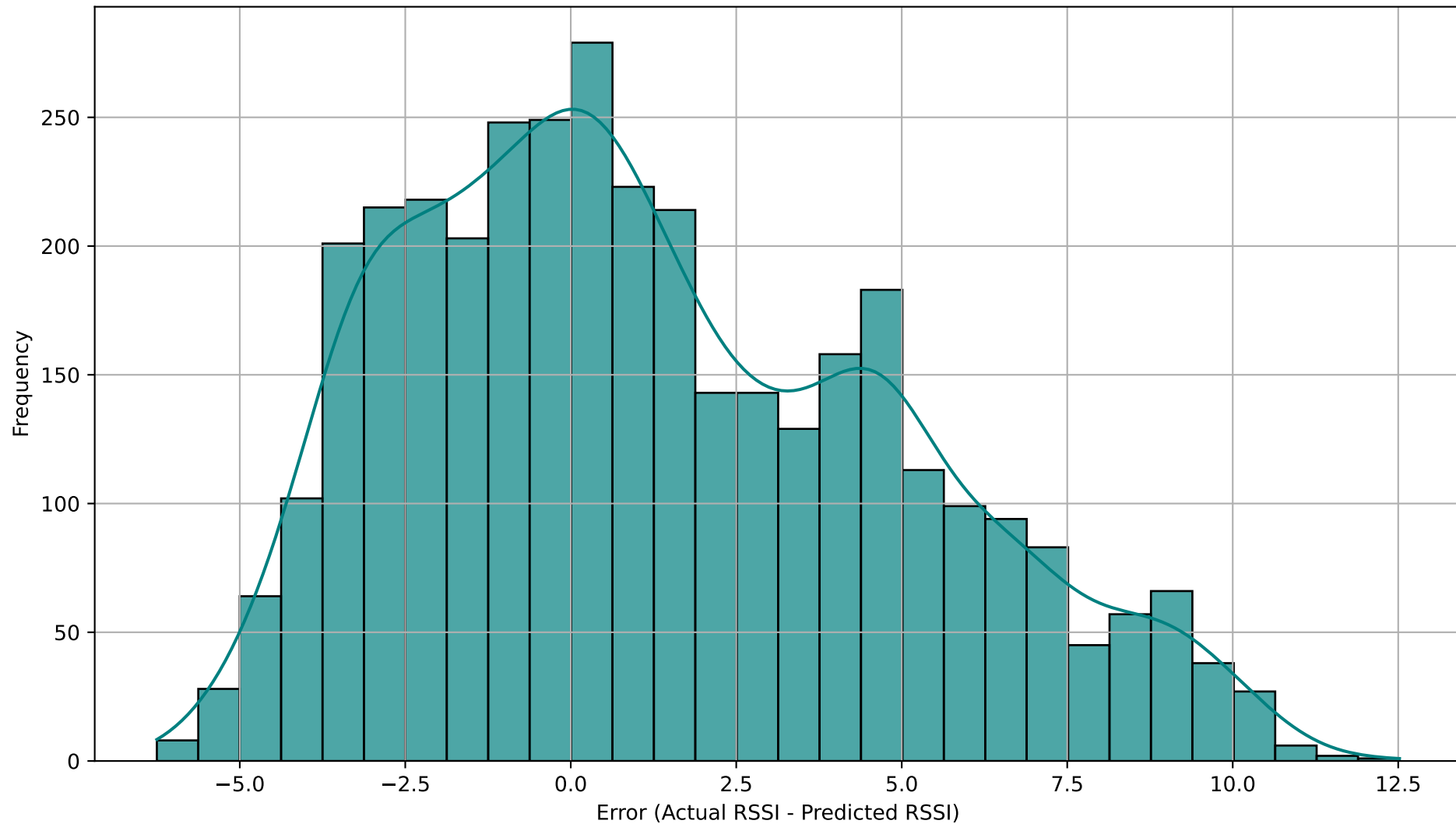
Actual vs Predicted RSSI (Last Step - step 40  $\approx$  4.0 s)



Distribution of Prediction Errors (First Step)

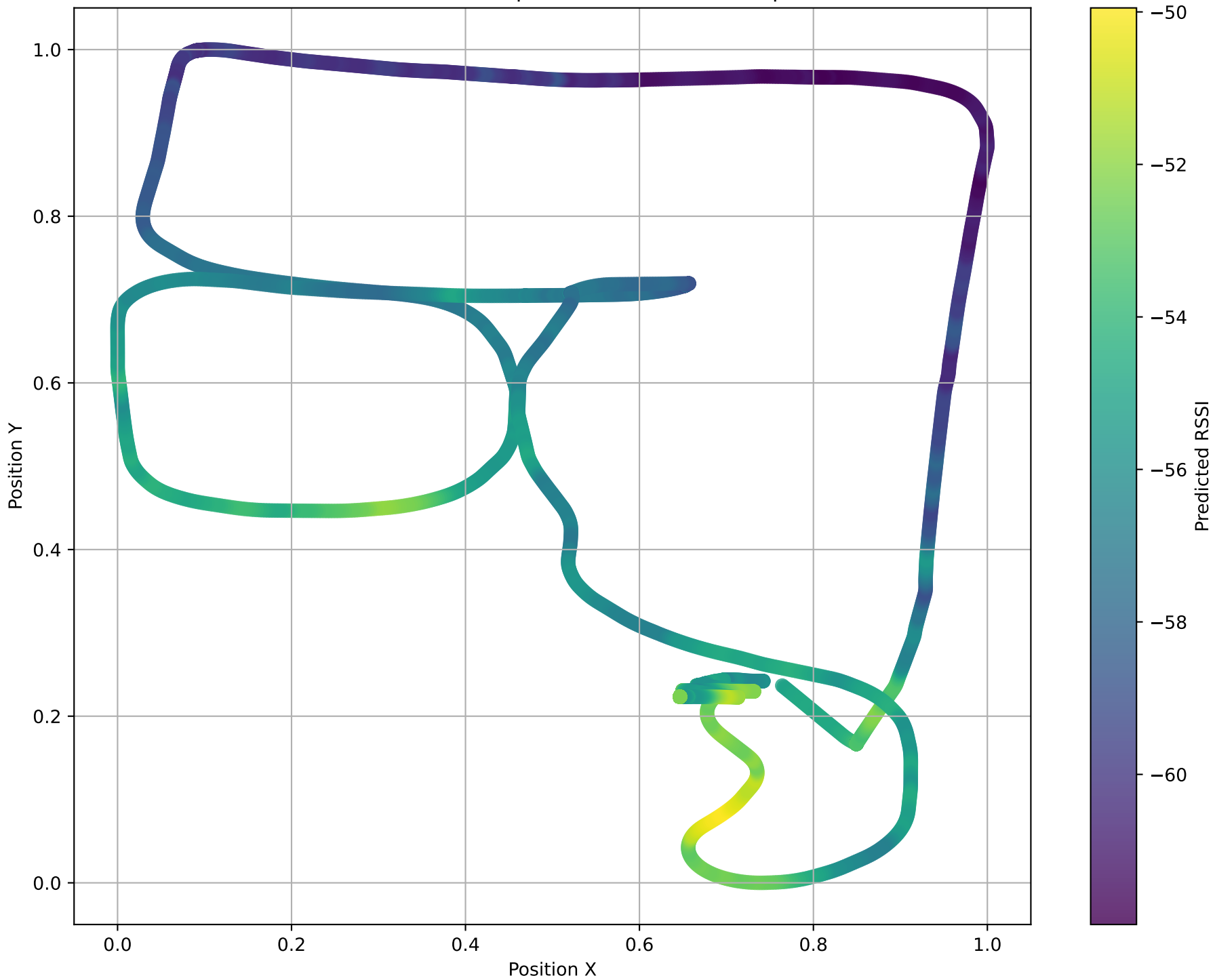


Distribution of Prediction Errors (Last Step - step 40  $\approx$  4.0 s)

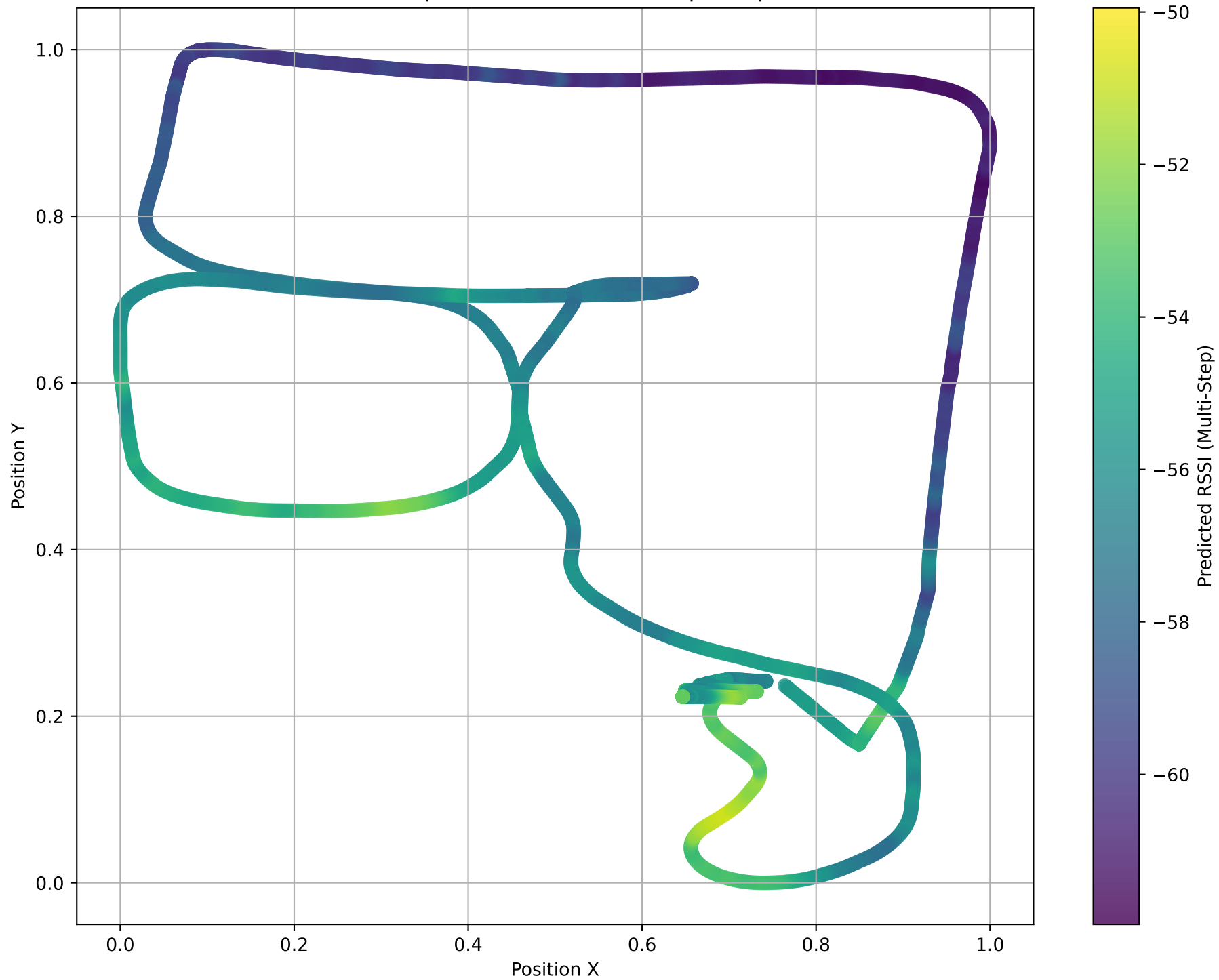




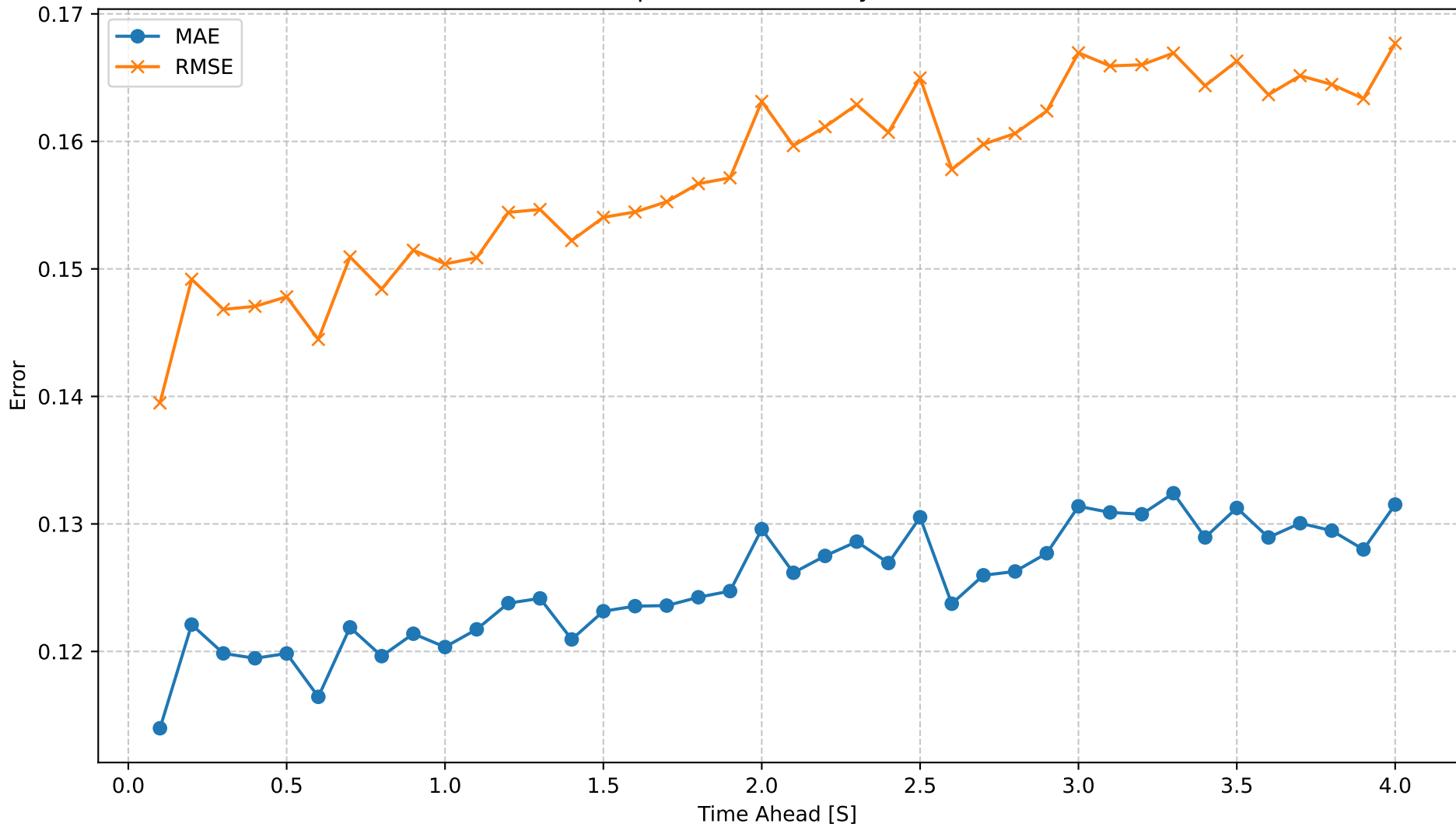
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40  $\approx$  4.0 s)



Multi-Step Forecast Error by Time Ahead



## Resumo Automático

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Dataset:            Training: rssi\_odom\_dataset\_11apr25\_isolado.parquet |  
Experiment: rssi\_odom\_dataset\_12apr25\_isolado.parquet  
Modelo:            LSTM  
Forecast steps: 40

### MAE / RMSE (dB)

- 1º passo : 0.11 / 0.14
- Mediana : 0.13 / 0.16
- 40º passo : 0.13 / 0.17

Crescimento do erro: 15.4% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 33 (0.13), 40 (0.13), 30 (0.13)

### Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.